

# Algorithm Cheat Sheet — All Templates + Java Syntax

## Conventions, Skeleton & Core API

### Generic Test Harness

```
class Question {
    // — PROBLEM-SOLVING PROCESS —————
    // 1. INPUT / OUTPUT : types, sizes, ranges; what to return; examples.
    // 2. HIGH-LEVEL ALGO : pick the pattern (two-ptr / BFS / DP / ...); state the idea in 1 line.
    // 3. IMPLEMENTATION : data structures, loop/recursion, invariants.
    // 4. EDGE CASES : empty, size 0/1, duplicates, negatives, overflow, all-same, not-found.
    // 5. COMPLEXITY : time & space; is it within limits?
    // 6. OPTIMIZATION : reduce time/space (hashing, two-pointer, rolling array, pruning, early exit)
    // —————
    private int data;
    public Question(int data) {
        this.data = data;
    }
    public void solve() {
        // TODO
    }
}

public class Solution {
    public static void main(String[] args) {
        Question q = new Question(0);
        q.solve();
    }
}
```

### Naming Conventions (enforced across all template files) Common Java API Cheat-Sheet

```
Map<Integer, Integer> map = new HashMap<>(); // key → value
Map<Integer, List<Integer>> adj = new HashMap<>(); // adjacency / buckets
Set<Integer> set = new HashSet<>(); // membership
List<Integer> list = new ArrayList<>(); // dynamic array
Deque<Integer> stack = new ArrayDeque<>(); // STACK: push / pop / peek
Deque<Integer> queue = new ArrayDeque<>(); // QUEUE: offer / poll / peek
// (ArrayDeque beats java.util.Stack / LinkedList)

PriorityQueue<Integer> minHeap = new PriorityQueue<>(); // min-heap (default)
PriorityQueue<Integer> maxHeap = new PriorityQueue<>(Collections.reverseOrder()); // max-heap
PriorityQueue<int[]> pq = new PriorityQueue<>((a, b) -> a[1] - b[1]); // by field, min-first
TreeMap<Integer, Integer> tmap = new TreeMap<>(); // sorted keys: floor/ceiling/first/last
TreeSet<Integer> tset = new TreeSet<>(); // sorted set: floor/ceiling/higher/lower
LinkedHashSet<Integer> lset = new LinkedHashSet<>(); // insertion-ordered set (LFU buckets)
Random rand = new Random(); // rand.nextInt(n) → 0..n-1 (RandomizedSet)
int[] arr = new int[n]; // defaults to 0
boolean[] seen = new boolean[n]; // defaults to false
int[][] grid = new int[m][n]; // 2D, all 0
int[] dr = {1,-1,0,0}, dc = {0,0,1,-1}; // DOWN,UP,RIGHT,LEFT; iterate for(dir=0..3) i+dr[dir],
```

```
// — Map —
map.put(k, v);
map.getOrDefault(k, 0); // safe read with default
map.merge(k, 1, Integer::sum); // ← counting frequency (most common)
map.computeIfAbsent(k, x -> new ArrayList<>()).add(v); // ← build adjacency / buckets
map.putIfAbsent(k, v); // ← write only first occurrence (keep earliest in
map.containsKey(k); map.remove(k);
for (Map.Entry<Integer,Integer> e : map.entrySet()) { e.getKey(); e.getValue(); e.setValue(v); }
map.keySet(); map.values();
// — Set —
set.add(x); set.contains(x); set.remove(x);
// — List —
list.add(x); list.get(i); list.set(i, x);
list.remove(list.size() - 1); // ← pop in backtracking
list.size(); list.isEmpty(); list.addAll(other);
Collections.sort(list); // ascending
list.sort((a, b) -> b - a); // custom / descending
Collections.reverse(list); // ← reverse path after building it backwards
```

```
// Deque as STACK (LIFO) // Deque as QUEUE (FIFO)
stack.push(x); queue.offer(x);
int top = stack.pop(); int head = queue.poll();
int look = stack.peek(); int look = queue.peek();
stack.isEmpty(); queue.isEmpty();
// Deque BOTH ENDS – needed for the MONOTONE DEQUE (sliding-window max/min)
// and for addFirst-style level building (zigzag BFS).
deque.offerLast(x); deque.pollLast(); deque.peekLast(); // back (push/pop/peek tail)
deque.offerFirst(x); deque.pollFirst(); deque.peekFirst(); // front (push/pop/peek head)
deque.addLast(x); deque.addFirst(x); // same as offer*, throw if capacity-bou
// PriorityQueue (heap)
pq.offer(x); int best = pq.poll(); int look = pq.peek(); pq.size();
pq.remove(x); // ← O(n) removal of an arbitrary element (lazy-deletion alternative in #480)
```

```
Integer.MAX_VALUE; // 2147483647 (2^31 - 1) – sentinel for "min so far"
Integer.MIN_VALUE; // -2147483648 (-2^31) – sentinel for "max so far"
Long.MAX_VALUE; // when sums can overflow int, accumulate in long
int k = i + (j - i) / 2; // ← avoid (i + j) overflow in binary search
Integer.compare(a, b); // ← overflow-safe comparator; use instead of (a - b)
(a, b) -> Integer.compare(a[1], b[1]); // ← safe PriorityQueue/Arrays.sort comparator
Double.compare(a, b); // ← comparator for double[] heaps (probability, median)
```

```
ch - 'a'; // 'a'..'z' → 0..25 (lowercase bucket index)
ch - '0'; // '0'..'9' → 0..9 (digit value)
(char) ('a' + i); // 0..25 → 'a'..'z'
num = num * 10 + (ch - '0'); // ← build a number digit by digit
Integer.parseInt("123"); // String → int (throws on bad input)
Integer.valueOf(123); // int → Integer (boxed)
String.valueOf(123); // int/char/etc → String
Character.getNumericValue('7'); // char → 7
// char classification
Character.isDigit(ch); Character.isLetter(ch); Character.isLetterOrDigit(ch);
Character.isWhitespace(ch); Character.toLowerCase(ch); Character.toUpperCase(ch);
```

```
s.length(); s.charAt(i); s.substring(i, j);          // [i, j)
s.toCharArray(); s.split(" "); s.equals(t); s.compareTo(t);
s.indexOf("ab"); s.isEmpty();
s.startsWith("ab"); s.endsWith("z"); s.contains("x"); // prefix / suffix / substring tests
s.toLowerCase(); s.toUpperCase();                  // case-normalize (e.g. case-insensitive compare)
s.trim(); s.replace('a', 'b'); s.repeat(n);         // strip ends / swap chars / repeat
StringBuilder builder = new StringBuilder();
builder.append(x); builder.reverse(); builder.toString();
builder.charAt(i); builder.setCharAt(i, c); builder.deleteCharAt(i); builder.length();
String.join(",", list);                            // list → "a,b,c"
```

```
Arrays.sort(arr);                                // ascending in place
Arrays.sort(intervals, (a, b) -> a[0] - b[0]);    // 2D by first field
Arrays.fill(dp, Integer.MAX_VALUE);              // seed a dp/dist array
Arrays.equals(window, need);                     // ← element-wise array compare (anagram check)
Arrays.copyOf(arr, len); Arrays.copyOfRange(arr, i, j);
System.arraycopy(src, srcPos, dst, dstPos, len); // ← fast block copy (merge step)
Arrays.asList(1, 2, 3);                          // fixed-size List view
int sum = Arrays.stream(arr).sum();
int max = Arrays.stream(arr).max().getAsInt();
// List<Integer> ↔ int[]
int[] a = list.stream().mapToInt(Integer::intValue).toArray();
List<Integer> l = Arrays.stream(a).boxed().collect(Collectors.toList());
```

```
Math.max(a, b); Math.min(a, b); Math.abs(x);
Math.pow(a, b); Math.sqrt(x); Math.ceil(x); Math.floor(x);
n & (n - 1);           // clear lowest set bit
n & (-n);              // isolate lowest set bit
1 << i;                // bit i set (use 1L << i for i ≥ 31)
(n >> i) & 1;          // read bit i
(n & 1) == 0;          // ← even (last bit 0)
(n & 1) == 1;          // ← odd (last bit 1)
n >> 1;                // ← divide by 2 (drop last bit); n << 1 = multiply by 2
Integer.bitCount(n);   // # of set bits
Integer.toBinaryString(n);
```

```
tmap.firstKey(); tmap.lastKey();
tmap.floorKey(x); // largest key ≤ x      tmap.ceilingKey(x); // smallest key ≥ x
tmap.lowerKey(x); // largest key < x      tmap.higherKey(x); // smallest key > x
tset.floor(x); tset.ceiling(x); tset.lower(x); tset.higher(x); tset.first(); tset.last();
```

```
// — NEGATIVE-SAFE MODULO — replaces the manual ((x % n) + n) % n
Math.floorMod(x, n); // always in [0, n-1] even when x < 0 (prefix-sum % k)
// — int[] INSTEAD OF HashMap for a bounded key space —
int[] count = new int[26]; // 0(1) no-boxing freq vs HashMap<Character,Integer>
count[ch - 'a']++; // also faster to compare: Arrays.equals(a, b)
// — 1D ROLLING ARRAY for DP — drop a dimension when row i only needs row i-1
int[] dp = new int[n + 1]; // 0(n) space instead of 0(m·n)
for (...) for (int j = target; j >= w; j--) dp[j] += dp[j - w]; // 0/1 knapsack: iterate j DOWN
// — FIXED-SIZE HEAP for top-k — 0(n log k) beats sorting 0(n log n)
if (heap.size() > k) heap.poll(); // keep only k best; min-heap for "k largest"
// — LAZY DELETION over pq.remove(obj) — pq.remove is 0(n); skip stale entries instead
if (entry[0] != dist[node]) continue; // Dijkstra: ignore outdated heap entries
// — ArrayDeque over Stack / LinkedList — faster, no synchronization, both ends 0(1)
Deque<Integer> stack = new ArrayDeque<>(); // never use java.util.Stack
// — StringBuilder over String += in a loop — 0(n) vs 0(n²)
StringBuilder builder = new StringBuilder(); // append, then builder.toString() once
// — EARLY EXIT / PRUNING — return the moment the answer is decided
if (found) return result; // e.g. backtracking bounds
```

## Two Pointers

### Pattern 1 — Opposite Two Pointers

```
int i = 0, j = n - 1;
while (i < j) {
    if (condition == target) {
        /* record */ i++; j--;
    } else if (tooSmall) {
        i++;
    } else {
        j--;
    }
}
```

## Sliding Window

### The Three Templates

```
int i = 0;
for (int j = 0; j < n; j++) {
    // 1. add nums[j] to state
    if (j - i + 1 == k) {
        // 2. record (window is exactly size k)
        // 3. remove nums[i] from state
        i++;
    }
}

int i = 0, result = 0;
for (int j = 0; j < n; j++) {
    // 1. add nums[j] to state
    while (/* window exceeds constraint */) {
        i++;
    }
    result = Math.max(result, j - i + 1);
}

int i = 0, result = Integer.MAX_VALUE;
for (int j = 0; j < n; j++) {
    // 1. add nums[j] to state
    while (/* window VALID */) {
        result = Math.min(result, j - i + 1);
        i++;
    }
}
```

### Frequency Map Trick (used by 567 and 76)

```
if (need[s.charAt(j)]-- > 0) required--;
if (need[s.charAt(i)]++ >= 0) required++;
i++;
```

## Prefix Sum + HashMap

### Core Template

```

Map<Long, Integer> map = new HashMap<>();
map.put(0L, ???);
long sum = 0;
for (int i = 0; i < nums.length; i++) {
    sum += nums[i];
    long lookup = transform(sum);
    map.put(storeKey(sum), storeValue(i)); // what to store (sum or sum%k → count or index)
}

```

## Binary Search

The Only Template — [i, j) , find first k with condition(k)

```

int i = 0, j = nums.length;
while (i < j) {
    int k = i + (j - i) / 2;
    if (!condition(k)) {
        i = k + 1;
    } else {
        j = k;
    }
}

```

How to Choose

## Sorting

Template 1 — Merge Sort

```

public void mergeSort(int[] nums) {
    mergeSort(nums, 0, nums.length, new int[nums.length]);
}
private void mergeSort(int[] nums, int start, int end, int[] temp) {
    if (end - start <= 1) return;
    int mid = start + (end - start) / 2;
    mergeSort(nums, start, mid, temp);
    mergeSort(nums, mid, end, temp);
    merge(nums, start, mid, end, temp);
}
private void merge(int[] nums, int start, int mid, int end, int[] temp) {
    int i = start, j = mid, k = start;
    while (i < mid || j < end) {
        if (j >= end || (i < mid && nums[i] <= nums[j])) {
            temp[k++] = nums[i++];
        } else {
            temp[k++] = nums[j++];
        }
    }
    for (i = start; i < end; i++) nums[i] = temp[i];
}

```

Template 2 — Quick Sort (3-way Dutch Flag Partition)

```

public void quickSort(int[] nums) {
    quickSort(nums, 0, nums.length);
}
private void quickSort(int[] nums, int start, int end) {
    if (end - start <= 1) return;
    int pivot = nums[start + (end - start) / 2];
    int i = start, k = start, j = end - 1;
    while (k <= j) {
        if (nums[k] < pivot) {
            swap(nums, k++, i++);
        } else if (nums[k] == pivot) {
            k++;
        } else {
            swap(nums, k, j--);
        }
    }
    quickSort(nums, start, i);
    quickSort(nums, k, end);
}
private void swap(int[] nums, int i, int j) {
    int t = nums[i]; nums[i] = nums[j]; nums[j] = t;
}

```

### Template 3 — Quick Select (partial sort — k-th element)

```

private int quickSelect(int[] nums, int start, int end, int target) {
    int pivot = nums[start + (end - start) / 2];
    int i = start, k = start, j = end - 1;
    while (k <= j) {
        if (nums[k] < pivot) {
            swap(nums, k++, i++);
        } else if (nums[k] == pivot) {
            k++;
        } else {
            swap(nums, k, j--);
        }
    }
    if (target < i) {
        return quickSelect(nums, start, i, target);
    } else if (target < k) {
        return pivot;
    } else {
        return quickSelect(nums, k, end, target);
    }
}

```

### Template 4 — Counting Sort

```

public void countingSort(int[] arr) {
    if (arr.length == 0) return;
    int min = Arrays.stream(arr).min().getAsInt();
    int max = Arrays.stream(arr).max().getAsInt();
    int[] buckets = new int[max - min + 1];
    for (int x : arr) buckets[x - min]++;
    int i = 0;
    for (int v = 0; v < buckets.length; v++)
        while (buckets[v]-- > 0) arr[i++] = v + min;
}

```

## Linked List

When to Use — Signal → Pattern All Four Templates

```

// 1. DELETE / FILTER - sentinel guards head removal; previous tracks last kept node
ListNode sentinel = new ListNode(0);
sentinel.next = head;
ListNode previous = sentinel, current = head;
while (current != null) {
    if (shouldDelete(current)) {
        previous.next = current.next;
    } else {
        previous = current;
    }
    current = current.next;
}
return sentinel.next;
// 2. REVERSAL - re-wire one node at a time
ListNode previous = null, current = head;
while (current != null) {
    ListNode next = current.next;
    current.next = previous;
    previous = current;
    current = next;
}
return previous;
// 3. FAST / SLOW - two pointers at different speeds
ListNode slow = head, fast = head;
while (fast != null && fast.next != null) {
    slow = slow.next;
    fast = fast.next.next;
}
// 4. MERGE - sentinel collects nodes from two lists in order
ListNode sentinel = new ListNode(0);
ListNode current = sentinel;
while (a != null && b != null) {
    if (a.val <= b.val) {
        current.next = a;
        a = a.next;
    } else {
        current.next = b;
        b = b.next;
    }
    current = current.next;
}
current.next = (a != null) ? a : b;
return sentinel.next;

```

## String

### Core Idioms

```

// 1. CHAR COUNT – lowercase letters
int[] count = new int[26];
for (char ch : s.toCharArray()) count[ch - 'a']++;
// 2. CHAR COUNT – digits 0–9
int[] count = new int[10];
for (char ch : s.toCharArray()) count[ch - '0']++;
// 3. CHAR TO INTEGER – build number digit by digit
int num = 0;
for (int i = 0; i < s.length(); i++)
    num = num * 10 + (s.charAt(i) - '0');
// 4. ANAGRAM HASH KEY – frequency-based (bucket sort style) O(k) vs sort O(k log k)
String hashCode(String s) {
    int[] count = new int[26];
    for (char ch : s.toCharArray()) count[ch - 'a']++;
    StringBuilder builder = new StringBuilder();
    for (int i = 0; i < 26; i++) {
        builder.append((char)('a' + i));
        builder.append(count[i]);
    }
    return builder.toString();
}
char[] chars = s.toCharArray();
Arrays.sort(chars);
String key = new String(chars);
// 5. BUCKET SORT – sort by frequency without comparison sort
int[] count = new int[128];
for (char ch : s.toCharArray()) count[ch]++;
List<Character>[] buckets = new List[s.length() + 1]; // index = frequency
for (int i = 0; i < 128; i++) {
    if (count[i] > 0) {
        int freq = count[i];
        if (buckets[freq] == null) buckets[freq] = new ArrayList<>();
        buckets[freq].add((char) i);
    }
}
StringBuilder builder = new StringBuilder();
for (int freq = buckets.length - 1; freq > 0; freq--) {
    if (buckets[freq] == null) continue;
    for (char ch : buckets[freq])
        for (int i = 0; i < freq; i++) builder.append(ch);
}
// 6. EXPAND FROM CENTER – palindrome check in O(1) space
int expand(String s, int i, int j) {
    while (i >= 0 && j < s.length() && s.charAt(i) == s.charAt(j)) { i--; j++; }
    return j - i - 1;
}

```

## Stack / Queue / Heap

### All Templates

```

// 1. STACK (LIFO) – use ArrayDeque, not Stack class
Deque<Integer> stack = new ArrayDeque<>();
stack.push(x);
stack.pop();
stack.peek();

// 2A. MONOTONE DECREASING STACK – next GREATER element
Deque<Integer> stack = new ArrayDeque<>();
for (int i = 0; i < n; i++) {
    while (!stack.isEmpty() && nums[stack.peek()] < nums[i])
        result[stack.pop()] = nums[i];
    stack.push(i);
}

// 2B. MONOTONE INCREASING STACK – next SMALLER element / span width
Deque<Integer> stack = new ArrayDeque<>();
for (int i = 0; i < n; i++) {
    while (!stack.isEmpty() && nums[stack.peek()] > nums[i]) {
        int height = nums[stack.pop()];
        int width = stack.isEmpty() ? i : i - stack.peek() - 1;
    }
    stack.push(i);
}

// 3. MONOTONE DEQUE – sliding window max/min
Deque<Integer> queue = new ArrayDeque<>();
for (int i = 0; i < n; i++) {
    while (!queue.isEmpty() && nums[queue.peekLast()] <= nums[i])
        queue.pollLast();
    queue.offerLast(i);
    if (queue.peekFirst() < i - k + 1) queue.pollFirst();
    if (i >= k - 1) result[i - k + 1] = nums[queue.peekFirst()];
}

// 4. PRIORITY QUEUE
PriorityQueue<Integer> minPQ = new PriorityQueue<>();
PriorityQueue<Integer> maxPQ = new PriorityQueue<>(Collections.reverseOrder()); // max-heap
PriorityQueue<int[]> pq = new PriorityQueue<>((a, b) -> a[1] - b[1]); // custom: sort by index 1
pq.offer(x);
pq.poll();
pq.peek();

// 5. TWO HEAPS – maintain median in O(log n)
PriorityQueue<Integer> maxHeap = new PriorityQueue<>(Collections.reverseOrder());
PriorityQueue<Integer> minHeap = new PriorityQueue<>();
void addNum(int num) {
    maxHeap.offer(num);
    minHeap.offer(maxHeap.poll());
    if (maxHeap.size() < minHeap.size())
        maxHeap.offer(minHeap.poll());
}
double findMedian() {
    return maxHeap.size() > minHeap.size()
        ? maxHeap.peek()
        : (maxHeap.peek() + minHeap.peek()) / 2.0;
}

```

## BFS (Tree)

### The Template — Level-Aware BFS

```

Queue<TreeNode> queue = new ArrayDeque<>();
queue.offer(root);
int level = 0;
while (!queue.isEmpty()) {
    int size = queue.size();
    level++;
    for (int i = 0; i < size; i++) {
        TreeNode node = queue.poll();
        if (node.left != null) queue.offer(node.left);
        if (node.right != null) queue.offer(node.right);
    }
}

```

### When to Use — Signal → Pattern The One Rule — Snapshot Level Size First

Always snapshot the level size BEFORE the inner `for` loop:

```

int size = queue.size();
for (int i = 0; i < size; i++) { ... }

```

Without the snapshot, newly added children mix with the current level and `i == size-1` / `i == 0` checks become meaningless.

## DFS / Backtracking

### When to Use — Signal → Pattern All Templates

```

// 1. PROCESS CHILDREN FIRST, COMBINE AT NODE (post-order)
private int dfs(TreeNode node) {
    if (node == null) return BASE;
    int left = dfs(node.left);
    int right = dfs(node.right);
    return ...;
}

// 2. PROCESS NODE FIRST, PASS STATE DOWN (pre-order)
private void dfs(TreeNode node, int accumulated) {
    if (node == null) return;
    accumulated += node.val;
    if (isLeaf(node)) { /* record */ return; }
    dfs(node.left, accumulated);
    dfs(node.right, accumulated);
}

// 3. TREE - GLOBAL ANSWER (return up the best single arm; update global across both arms)
int answer = Integer.MIN_VALUE;
private int dfs(TreeNode node) {
    if (node == null) return 0;
    int left = Math.max(0, dfs(node.left));
    int right = Math.max(0, dfs(node.right));
    answer = Math.max(answer, left + right + node.val);
    return Math.max(left, right) + node.val;
}

// 4. GRID DFS - mark visited by mutating grid; 4-directional flood fill
private void dfs(int[][] grid, int r, int c) {
    if (r < 0 || r >= grid.length || c < 0 || c >= grid[0].length) return;
    if (grid[r][c] != TARGET) return;
    grid[r][c] = VISITED;
    dfs(grid, r+1, c); dfs(grid, r-1, c);
    dfs(grid, r, c+1); dfs(grid, r, c-1);
}

// 5. GRAPH DFS - 3-color visited: 0=unseen 1=in-stack 2=done
int[] state;
private boolean dfs(int node) {
    if (state[node] == 1) return true;
    if (state[node] == 2) return false;
    state[node] = 1;
    for (int neighbor : graph.get(node))
        if (dfs(neighbor)) return true;
    state[node] = 2;
    return false;
}

// 6. BACKTRACKING - choose / recurse / undo
private void backtrack(int start, List<Integer> current) {
    if (baseCase) { result.add(new ArrayList<>(current)); return; }
    for (int i = start; i < n; i++) {
        if (skipCondition(i)) continue;
        current.add(nums[i]);
        backtrack(next, current);
        current.remove(current.size()-1);
    }
}
}

```

## Graph

### Complexity Legend

|                  |                                          |                                                    |
|------------------|------------------------------------------|----------------------------------------------------|
| BFS / DFS        | $O(V + E)$                               | visit each vertex and edge once                    |
| Topo sort (Kahn) | $O(V + E)$                               | each node enqueued once, each edge relaxed once    |
| Union-Find       | $\sim O(n \cdot \alpha(n)) \approx O(n)$ | $\alpha$ = inverse Ackermann, effectively constant |
| Dijkstra         | $O((V + E) \log V)$                      | non-negative weights ONLY                          |

## Graph Representation

```
List<List<Integer>> graph = new ArrayList<>();
for (int i = 0; i < n; i++) graph.add(new ArrayList<>());
for (int[] edge : edges) {
    graph.get(edge[0]).add(edge[1]);
    graph.get(edge[1]).add(edge[0]);
}

List<List<int[]>> graph = new ArrayList<>();
for (int i = 0; i < n; i++) graph.add(new ArrayList<>());
for (int[] edge : edges)
    graph.get(edge[0]).add(new int[]{edge[1], edge[2]});
int[] dr = {1, -1, 0, 0};
int[] dc = {0, 0, 1, -1};
```

## Core Templates

```

// 1. BFS – shortest path / level order (track distance by level-size snapshot)
Queue<Integer> queue = new ArrayDeque<>();
boolean[] visited = new boolean[n];
queue.offer(start);
visited[start] = true;
int level = 0;
while (!queue.isEmpty()) {
    int size = queue.size();
    for (int i = 0; i < size; i++) {
        int node = queue.poll();
        for (int neighbor : graph.get(node)) {
            if (!visited[neighbor]) {
                visited[neighbor] = true;
                queue.offer(neighbor);
            }
        }
    }
    level++;
}

// 2. TOPOLOGICAL SORT – Kahn's BFS
int[] inDegree = new int[n];
for (int[] edge : edges) inDegree[edge[1]]++;
Queue<Integer> queue = new ArrayDeque<>();
for (int i = 0; i < n; i++) if (inDegree[i] == 0) queue.offer(i);
List<Integer> order = new ArrayList<>();
while (!queue.isEmpty()) {
    int node = queue.poll();
    order.add(node);
    for (int neighbor : graph.get(node))
        if (--inDegree[neighbor] == 0) queue.offer(neighbor);
}

// 3. UNION FIND
int[] parent, rank;
void init(int n) {
    parent = new int[n]; rank = new int[n];
    for (int i = 0; i < n; i++) parent[i] = i;
}
int find(int x) {
    if (parent[x] != x) parent[x] = find(parent[x]);
    return parent[x];
}
boolean union(int x, int y) {
    int px = find(x), py = find(y);
    if (px == py) return false;
    if (rank[px] < rank[py]) { int t = px; px = py; py = t; }
    parent[py] = px;
    if (rank[px] == rank[py]) rank[px]++;
    return true;
}

// 4. DIJKSTRA – shortest path (non-negative weights)
int[] dist = new int[n];
Arrays.fill(dist, Integer.MAX_VALUE);
dist[src] = 0;
PriorityQueue<int[]> pq = new PriorityQueue<>((a, b) -> a[1] - b[1]);
pq.offer(new int[]{src, 0});
while (!pq.isEmpty()) {
    int[] current = pq.poll();
    int node = current[0], d = current[1];
    if (d > dist[node]) continue;
    for (int[] nb : graph.get(node)) {
        int next = nb[0], w = nb[1];
        if (dist[node] + w < dist[next]) {
            dist[next] = dist[node] + w;
        }
    }
}

```

```

        pq.offer(new int[]{next, dist[next]});
    }
}
// 5. BIPARTITE CHECK – BFS 2-coloring
int[] color = new int[n];
Arrays.fill(color, -1);
Queue<Integer> queue = new ArrayDeque<>();
for (int start = 0; start < n; start++) {
    if (color[start] != -1) continue;
    color[start] = 0;
    queue.offer(start);
    while (!queue.isEmpty()) {
        int node = queue.poll();
        for (int neighbor : graph.get(node)) {
            if (color[neighbor] == -1) { color[neighbor] = 1 - color[node]; queue.offer(neighbor); }
            else if (color[neighbor] == color[node]) return false;
        }
    }
}
return true;

```

## Greedy

### Two Interval Templates

MERGE DIRECTION RULE (for merge-style problems like #56, where both keys work):  
 Sort by START → traverse FORWARD (i = 0 → n-1), extend the END of last kept  
 Sort by END → traverse BACKWARD (i = n-1 → 0), extend the START of last kept  
 These two are exact mirror images. See #56 for both written out side by side.  
 (NOTE: this mirror only applies to merging. Template 1 below sorts by end but still sweeps FORWARD – there the goal is counting, not merging.)

TEMPLATE 1 – Sort by END time, sweep forward, track end of last kept interval  
 Use when: maximizing count of non-overlapping intervals  
 Greedy insight: earliest-finishing interval leaves most room for future ones

TEMPLATE 2 – Sort by START time, sweep forward, merge when overlapping  
 Use when: merging/counting overlapping intervals  
 Greedy insight: processing in start order → each interval only needs to compare with the last merged result

```

Arrays.sort(intervals, (a, b) -> a[1] - b[1]);
int lastEnd = Integer.MIN_VALUE;
for (int[] interval : intervals) {
    if (interval[0] >= lastEnd) {
        lastEnd = interval[1];
    } else {
    }
}
Arrays.sort(intervals, (a, b) -> a[0] - b[0]);
List<int[]> result = new ArrayList<>();
for (int[] interval : intervals) {
    if (result.isEmpty() || result.get(result.size()-1)[1] < interval[0]) {
        result.add(interval);
    } else {
        result.get(result.size()-1)[1] =
            Math.max(result.get(result.size()-1)[1], interval[1]);
    }
}
Arrays.sort(intervals, (a, b) -> a[0] - b[0]);
PriorityQueue<Integer> pq = new PriorityQueue<>();
for (int[] interval : intervals) {
    if (!pq.isEmpty() && pq.peek() <= interval[0])
        pq.poll();
    pq.offer(interval[1]);
}

```

## Dynamic Programming

### All Six Templates

```

// 1. LINEAR 1D – each cell depends on a fixed number of previous cells
int[] dp = new int[n + 1];
dp[0] = base;
for (int i = 1; i <= n; i++)
    dp[i] = f(dp[i-1], dp[i-2], ...);
// 2A. LOOK-BACK 1D – dp[i] depends on all j < i
int[] dp = new int[n];
for (int i = 0; i < n; i++) {
    for (int j = 0; j < i; j++) {
        dp[i] = f(dp[i], dp[j]);
    }
}
// 3. 2D SEQUENCE – match consumes both, mismatch drops one side. WHEN: LCS / edit
// 4. INTERVAL DP – short ranges first (i right-to-left, j from i+1; dp[i+1][*] ready).
// 5. GRID DP – each cell accumulates from cells it could arrive from (up/left).
// 6. STATE MACHINE – named states updated from yesterday's states each step.

```

Want count or min/max?

```

├─ Count (dp[j] += dp[j - num])
│   ├── 0/1 (each item once):    j descending → #416, #494
│   └─ Unbounded (reuse):        j ascending  → #518
│       └─ Ordered (permut):     target outer, nums inner → #377
└─ Min/Max (dp[j] = min/max(...))
    ├── 0/1 minimize:            j descending → #474 (maximize)
    └─ Unbounded minimize:       j ascending  → #322

```

```
// — 0/1 KNAPSACK — each item at most once
int[] dp = new int[target + 1];
dp[0] = 1;
for (int i = 0; i < nums.length; i++) {
    for (int j = target; j >= nums[i]; j--)
        dp[j] += dp[j - nums[i]];
}

// — UNBOUNDED KNAPSACK — each item reusable
int[] dp = new int[target + 1];
dp[0] = 1;
for (int i = 0; i < nums.length; i++) {
    for (int j = nums[i]; j <= target; j++)
        dp[j] += dp[j - nums[i]];
}

// — PERMUTATION COUNT — order matters (Combination Sum IV)
int[] dp = new int[target + 1];
dp[0] = 1;
for (int j = 1; j <= target; j++) {
    for (int num : nums) {
        if (j >= num) dp[j] += dp[j - num];
    }
}
}
```

```
int[][] dp = new int[m + 1][n + 1];
for (int i = 1; i <= m; i++) {
    for (int j = 1; j <= n; j++) {
        if (s1.charAt(i-1) == s2.charAt(j-1)) {
            dp[i][j] = dp[i-1][j-1] + 1;
        } else {
            dp[i][j] = combine(dp[i-1][j], dp[i][j-1], dp[i-1][j-1]);
        }
    }
}
}
```

```
int[][] dp = new int[n][n];
for (int i = 0; i < n; i++) dp[i][i] = base_case;
for (int i = n - 1; i >= 0; i--) {
    for (int j = i + 1; j < n; j++) {
        dp[i][j] = ...;
    }
}
}
```

```
int[][] dp = new int[n][n];
for (int i = 0; i < n; i++) dp[i][i] = base_case;
for (int i = 0; i < n; i++) {
    for (int j = i - 1; j >= 0; j--) {
        dp[i][j] = ...;
    }
}
}
```

```
int[][] dp = new int[n][n];
for (int i = 0; i < n; i++) dp[i][0] = 1;
for (int i = 1; i < n; i++) {
    for (int j = 1; j <= i; j++) {
        dp[i][j] = ...;
    }
}
}
```

```
int[] dr = {1,-1,0,0}, dc = {0,0,1,-1};
int[][] dp = new int[rows][cols];
for (int i = 0; i < rows; i++)
    for (int j = 0; j < cols; j++)
        dp[i][j] = grid[i][j] + f(dp[i-1][j], dp[i][j-1]);
int[][] memo = new int[rows][cols];
for (int i = 0; i < rows; i++)
    for (int j = 0; j < cols; j++)
        dfs(grid, i, j, memo, dr, dc);
```

cash = max profit **when NOT holding** (free to buy or idle)  
 hold = max profit **when HOLDING stock** (bought it at some cost)  
 cooldown = max profit right after **selling** (can't buy today)

|                   | buy                                | sell | re-buy condition        |
|-------------------|------------------------------------|------|-------------------------|
| #121 (1 tx):      | hold = max(hold, -price)           |      | no re-buy (ignore cash) |
| #122 (unlimited): | hold = max(hold, cash - price)     |      | re-buy with full cash   |
| #123 (2 tx):      | chain: buy2 uses sell1 cash        |      | 4 states chained        |
| #309 (cooldown):  | hold = max(hold, cooldown - price) |      | must wait 1 day         |
| #714 (fee):       | hold = max(hold, cash - price)     |      | pay fee on sell         |

## Bit Manipulation

### Canonical Template

```
// — XOR CANCEL — pairs cancel; lone element survives
int result = 0;
for (int num : nums) result ^= num;
// — BIT OPERATIONS —
boolean isSet = ((n >> i) & 1) == 1;
int set = n | (1 << i);
int clear = n & ~(1 << i);
int toggle = n ^ (1 << i);
int lowest = n & (-n);
boolean isPow2 = n > 0 && (n & (n-1)) == 0;
// — COUNT SET BITS — Brian Kernighan: each iteration removes one set bit
int count = 0;
while (n != 0) { count++; n &= (n - 1); }
// — BITMASK AS SET — 26-bit integer represents presence of 26 letters
int mask = 0;
for (char c : word.toCharArray()) mask |= (1 << (c - 'a'));
if ((mask[i] & mask[j]) == 0) { /* no shared letter */ }
```

### Variations

```

int ones = 0, twos = 0;
for (int num : nums) {
    ones = (ones ^ num) & ~twos;
    twos = (twos ^ num) & ~ones;
}
int xor = 0;
for (int num : nums) xor ^= num;
int diff = xor & (-xor);
int a = 0;
for (int num : nums)
    if ((num & diff) != 0) a ^= num;
int b = xor ^ a;
int[] dp = new int[n + 1];
for (int i = 1; i <= n; i++)
    dp[i] = dp[i >> 1] + (i & 1);
int shift = 0;
while (left != right) { left >>= 1; right >>= 1; shift++; }
return left << shift;
int add(int a, int b) {
    while (b != 0) {
        int carry = a & b;
        a = a ^ b;
        b = carry << 1;
    }
    return a;
}

```

## Design

### Canonical Template — HashMap + Doubly Linked List (LRU)

```

class CacheNode {
    int key, value;
    CacheNode previous, next;
    CacheNode(int key, int value) { this.key = key; this.value = value; }
}
// — SENTINEL NODES — head (MRU side) ↔ ... ↔ tail (LRU side)
CacheNode head = new CacheNode(0, 0), tail = new CacheNode(0, 0);
head.next = tail; tail.previous = head;
// — REMOVE NODE —
void remove(CacheNode node) {
    node.previous.next = node.next;
    node.next.previous = node.previous;
}
// — INSERT AFTER HEAD (= mark as MRU) —
void insertAfterHead(CacheNode node) {
    node.next = head.next;
    node.previous = head;
    head.next.previous = node;
    head.next = node;
}
// — LRU EVICTION — tail.previous is LRU
CacheNode lru = tail.previous;
remove(lru);
map.remove(lru.key);

```

### Variations

## Math

### Canonical Templates

```

// — DIGIT EXTRACTION LOOP — peel digits right-to-left
while (n != 0) {
    int digit = n % 10;
    n /= 10;
}

// — FAST POWER (exponentiation by squaring) —
double fastPow(double x, long n) {
    double result = 1;
    while (n > 0) {
        if ((n & 1) == 1) result *= x;
        x *= x;
        n >>= 1;
    }
    return result;
}

// — BASE CONVERSION — generic base b, 0-indexed
StringBuilder builder = new StringBuilder();
while (num > 0) {
    builder.append(num % b);
    num /= b;
}
builder.reverse();
int value = 0;
for (int digit : digits) value = value * b + digit;

// — SIEVE OF ERATOSTHENES — mark composites up to n
boolean[] composite = new boolean[n];
for (int i = 2; i * i < n; i++) {
    if (composite[i]) continue;
    for (int j = i * i; j < n; j += i)
        composite[j] = true;
}

```